For evaluation, we use full-length Rouge F1 using the official evaluation tool [2]. In their work, the authors of (Bahdanau et al., 2014) use full-length Rouge Recall on this corpus, since the maximum length of limited-length version of Rouge recall of 75 bytes (intended for DUC data) is already long for Gigaword summaries. However, since full-length Recall can unfairly reward longer summaries, we also use full-length F1 in our experiments for a fair comparison between our models, independent of the summary length.

The experimental results comparing the Pointer Softmax with NMT model are displayed in Table 1 for the *UNK* pointers data and in Table 2 for the entity pointers data. As our experiments show, pointer softmax improves over the baseline NMT on both UNK data and entities data. Our hope was that the improvement would be larger for the entities data since the incidence of pointers was much greater. However, it turns out this is not the case, and we suspect the main reason is anonymization of entities which removed data-sparsity by converting all entities to integer-ids that are shared across all documents. We believe that on de-anonymized data, our model could help more, since the issue of data-sparsity is more acute in this case.

Table 1: Results on Gigaword Corpus when pointers are used for UNKs in the training data, using Rouge-F1 as the evaluation metric.

	Rouge-1	Rouge-2	Rouge-L
NMT + lvt	34.87	16.54	32.27
NMT + lvt + PS	**35.19**	**16.66**	**32.51**

Table 2: Results on anonymized Gigaword Corpus when pointers are used for entities, using Rouge-F1 as the evaluation metric.

	Rouge-1	Rouge-2	Rouge-L
NMT + lvt	34.89	**16.78**	32.37
NMT + lvt + PS	**35.11**	16.76	**32.55**

In Table 3, we provide the results for summarization on Gigaword corpus in terms of recall as also similar comparison is done by (Rush et al., 2015). We observe improvements on all the scores with the addition of pointer softmax. Let us note that, since the test set of (Rush et al., 2015) is not publicly available, we sample 2000 texts with their summaries without replacement from the validation set and used those examples as our test set.

Table 3: Results on Gigaword Corpus for modeling UNK's with pointers in terms of recall.

	Rouge-1	Rouge-2	Rouge-L
NMT + lvt	36.45	17.41	33.90
NMT + lvt + PS	**37.29**	**17.75**	**34.70**

In Table 4 we present a few system generated summaries from the Pointer Softmax model trained on the *UNK* pointers data. From those examples, it is apparent that the model has learned to accurately point to the source positions whenever it needs to generate rare words in the summary.

5.3 Neural Machine Translation

In our neural machine translation (NMT) experiments, we train NMT models with attention over the Europarl corpus (Bahdanau et al., 2014) over the sequences of length up to 50 for English to French translation. [3]. All models are trained with early-stopping which is done based on the negative log-likelihood (NLL) on the development set. Our evaluations to report the performance of our models are done on `newstest2011` by using BLUE score. [4]

We use $30,000$ tokens for both the source and the target language shortlist vocabularies (1 of the token is still reserved for the unknown words). The whole corpus contains $134,831$ unique English words and $153,083$ unique French words. We have created a word-level dictionary from French to English which contains translation of 15,953 words that are neither in shortlist vocabulary nor dictionary of common words for both the source and the target. There are about $49,490$ words shared between English and French parallel corpora of Europarl.

During the training, in order to decide whether to pick a word from the source sentence using attention/pointers or to predict the word from the short-list vocabulary, we use the following simple heuristic. If the word is not in the short-list

[2] `http://www.berouge.com/Pages/default.aspx`

[3] In our experiments, we use an existing code, provided in `https://github.com/kyunghyuncho/dl4mt-material`, and on the original model we only changed the last softmax layer for our experiments

[4] We compute the BLEU score using the multi-blue.perl script from Moses on tokenized sentence pairs.